Research Statement

I am an applied microeconomist specializing in environmental, energy, and industrial organization economics. My research studies how market structure shapes the outcomes of energy and environmental policy: who captures the value of clean-energy subsidies, whether firms exercise market power when supply tightens, and whether new technology can offset the external costs that one person's choices impose on others. A central theme of my work is that the effect of a policy depends not only on its design, but also on the competitive conditions of the market through which it operates, and that measuring those conditions credibly requires detailed, unit-level data.

My job market paper, *Solar Pass-through*, asks how much of a production-based solar subsidy reaches the households that install solar panels, and what the answer reveals about competition among installers. Massachusetts offers an unusual setting for this question because it replaced one production-based subsidy with another. Under the Solar Carve-Out program (2014–2018), owners earned tradable Solar Renewable Energy Certificates (SRECs) sold at market prices; under its successor, SMART, they receive a fixed rate per kilowatt-hour set at the time of application. Using data on more than 39,000 residential systems installed between 2014 and 2023, I construct the present value of the subsidy for each individual roof. I merge technical specifications from Lawrence Berkeley National Laboratory's Tracking the Sun database with state regulatory records, and use NREL's PVWatts model to predict each system's annual generation from its exact location, size, orientation, and equipment. Because a production-based subsidy pays for each unit of electricity generated, two otherwise similar systems installed by the same installer, in the same town and in the same month, receive different subsidies when one roof faces south and the other faces east. This variation across roofs does not depend on the policy calendar, which allows me to separate pass-through from the falling installation costs that accompanied changes in subsidy levels over time.

Baseline estimates suggest that pass-through differs across the two programs: 76 percent under the Solar Carve-Out and 102 percent under SMART. Much of this gap, however, reflects how each subsidy moved over time alongside falling installation costs. When only the variation across roofs is used, pass-through is about 85 percent in both programs, and the two estimates are statistically indistinguishable. Households therefore capture most of a production-based subsidy, as they do with cash rebates, but not all of it.

I then use these estimates to learn about market power. Using the sufficient-statistics framework of Weyl and Fabinger (2013), Pless and van Benthem (2019) find more-than-complete pass-through and convex demand for leased systems in California and conclude that the market there is imperfectly competitive. I apply the same test in reverse. Estimating demand at the ZIP-code level, I find that demand under SMART is log-concave, so market power would push pass-through below one. High pass-through therefore bounds installer markups at roughly two to four percent of price at the point estimates. Reduced-form evidence agrees: pass-through does not fall when installers are busy and does not vary significantly with local installer concentration. These findings suggest that the Massachusetts market for host-owned systems is close to competitive.

Although there is a substantial body of research on solar subsidy pass-through, most of the evidence comes from California, relies on ZIP-code-level measures of solar resource, and studies upfront rebates rather than production-based payments. My job market paper contributes to this literature in three ways. First, it compares pass-through across two production-based subsidy designs within the same market, a comparison the literature has so far lacked. Second, it shows that measuring subsidies roof by roof matters: most of the variation in subsidy value lies within towns, so town or statewide averages discard the variation that identifies pass-through. Third, it shows that pass-

through can be used not only to detect market power but also to bound it. For policymakers, the results suggest that in a competitive installer market a production-based subsidy reaches buyers largely, though not fully, and that comparisons of subsidy designs should separate changes in the subsidy from the cost trends that accompany them.

My working paper, *Nuclear Outages*, studies market power on the supply side of the electricity sector. Wholesale electricity markets are especially vulnerable to market power because demand is nearly perfectly inelastic, storage is expensive, and capacity constraints bind. We use nuclear refueling outages in New England as a test. Nuclear power has a low marginal cost and sits at the bottom of the supply curve, but each reactor must be refueled roughly every 18 months, which typically takes it offline for about a month. The resulting shortfall in baseload supply gives the remaining suppliers a stronger incentive to raise their offers. Using day-ahead hourly market data from 2016 to 2018, we find that the market-clearing price rises by \$4.9/MWh on average when at least one of the region's four reactors is offline. Analyzing suppliers' bidding behavior, we find that approximately 36 percent of this price increase can be attributed to the exercise of market power, which implies a transfer from consumers to suppliers of about \$43 to \$45 million per year.

These findings carry implications beyond nuclear power. As the grid transitions toward intermittent renewable resources, the hours when the sun does not shine and the wind does not blow create similar temporary shortfalls in low-cost supply, and may give the remaining fossil-fuel generators the same opportunity to raise markups. Together with my job market paper, this project reflects my broader interest in how competition, in both retail technology markets and wholesale electricity markets, determines who bears the cost of the clean-energy transition.

My other project, *Vehicle Weight*, examines whether technology can offset a well-documented externality in transportation. Heavier vehicles protect their own occupants but impose risk on the occupants of the vehicles they hit; Anderson and Auffhammer (2014) show that being struck by a vehicle 1,000 pounds heavier raises fatality risk by 40 to 50 percent. Since then, vehicles have become both heavier and far more likely to be equipped with advanced driver-assistance systems such as forward collision warning and automatic emergency braking. Using about 785,000 vehicle-level records from Massachusetts crash reports between 2021 and 2025, matched to curb weight and driver-assistance equipment through VIN decoding, we first confirm the weight externality in a modern fleet: in two-vehicle crashes, holding a vehicle's own weight fixed, a 1,000-pound heavier collision partner raises the probability of severe or fatal injury by about 13 percent relative to the baseline rate. Preliminary results show that all seven driver-assistance features we examine are associated with a smaller weight externality, and that the association is statistically significant for dynamic brake support. Because these features are often bundled in trim packages and correlated with vehicle age and segment, our next step is to exploit the staggered introduction of specific features across model years to move from correlation toward causal estimates. If driver-assistance technology does mitigate the weight externality, safety-technology mandates could partially offset the external cost of a heavier, increasingly electric vehicle fleet.

Going forward, I plan to extend my research in two directions. First, I plan to study the impact of elevated electricity demand in the US on energy transitions. Second, I plan to study market power in electricity markets with growing renewable penetration, using hourly variation in wind and solar output as supply shocks analogous to nuclear outages, to test whether intermittency creates new opportunities for the exercise of market power and how market design can limit them.

References

[1] "Subsidy Pass-through and Market Power: Evidence from Production-Based Solar Subsidies" Xiaolin Zhou, Job Market Paper. Referred to as *Solar Pass-through*

[2] "Market Power in the New England Electricity Market: Evidence from Nuclear Refueling Outages" Xiaolin Zhou, Ming Ge, James Ji and Matt Woerman, Working Paper. Referred to as *Nuclear Outages*

[3] "Can Driver-Assistance Technology Offset the Vehicle-Weight Externality?" Xiaolin Zhou and Yongjoon Park, Working Paper. Referred to as *Vehicle Weight*